

Programmable Money and the Abolition of Visible Failure

A price ceiling meant to make bread affordable instead ensures its scarcity. A credit expansion meant to stimulate the macroeconomy ends in a burst bubble. A prohibition meant to suppress a trade merely relocates it out of the purview of government officials. Mises showed that these consequences are not accidents but the predictable result of obstructing the price system, whose distortions are almost always blamed on something other than the policy, so that each intervention prepares the ground for the next.¹ And thus, as the Austrian argument goes, the economy eventually arrives at its predestined terminus: the fully planned economy. By that point, the consequences of state intervention are no longer of any consequence.

What the story takes for granted is easy to overlook, because until now it has always held true: interventions fail in public. The shortage is visible, whether it be a physical queue or a black market. Therefore, although a failed intervention remains a substantial cost, it is nonetheless a prolific generator of information, i.e., price information. With this information, the state can either refashion it into its next pretext to justify further intervention or, less likely, cease its original intervention. A housing shortage caused by rent control, in other words, can either lead to the imposition of further rent stabilization laws to prevent the remaining units from raising their prices or, as the consequences of intervention are evident, motivate the state to terminate its original attempt at rent control.²

This dichotomy is rendered largely moot, however, with the rise of *programmable money*. Rather, programmable money, by which I mean *ex ante* state control of the protocol of payment, is a unique form of intervention that obfuscates the typically visible price signal. Intervention in the conventional sense forbids an action but then must be able to surveil and enforce its intervention, but programmable money invalidates the action before it occurs. The danger is therefore not that, in the age of programmable money, a state imposes stricter controls than before, but that such “failures” (if they can continue to be categorized as such, without the previously evident repercussions) now *fail* to produce their necessary byproduct, that being

¹Ludwig von Mises, *A Critique of Interventionism*, trans. Hans F. Sennholz (Auburn, AL: Ludwig von Mises Institute, 2011); and Ludwig von Mises, *Human Action: A Treatise on Economics*, Scholar’s ed. (Auburn, AL: Ludwig von Mises Institute, 1998).

²Robert L. Schuettinger and Eamonn F. Butler, *Forty Centuries of Wage and Price Controls: How Not to Fight Inflation* (Washington, DC: Heritage Foundation, 1979).

public evidence of an intervention's consequence. This is the crucial evidence on which both the market's self-correction and the state's eventual retreat have depended.

Mises demonstrated that socialism cannot calculate,³ but such a state can at least perceive the physical disorder its planning produces. But a state that programs its money *can no longer even learn that it can no longer calculate* because the very nature of that initial intervention blinds it to the result of subsequent interventions. So, if visible failure is what has checked intervention, then the institutions that keep failure in the public eye must be protected as the economy's error-detection apparatus.

Failure as Signal

Mises's argument against interventionism proceeds in two steps. First, an intervention fails to accomplish its intended goal: a price ceiling meant to keep bread affordable induces a shortage of it. Second, the failure then forces a choice between repeal and extension, where the politics of blame favor extension, since the visible distortion is attributed to speculators, hoarders, or foreign saboteurs rather than to the initial state action.⁴ Rationing ensues to combat the shortage, which then requires production controls to sustain it, and so on, until at the end stands the planned economy whose defining problem Mises had established elsewhere: without private property in the means of production, there are no genuine prices, and without prices, there is no economic calculation.

Hayek adds an epistemic dimension to the problem of economic calculation. The knowledge that coordination requires is dispersed, tacit, and local, with the only mechanism capable of harnessing that information found in the price system, which aggregates knowledge of the particular circumstances of time and place.⁵ Kirzner and Huerta de Soto expand the argument, illustrating how entrepreneurial knowledge is not merely diffused but *created* in the very act of price discovery: an entrepreneur who notices that a good can be produced more cheaply brings into existence a fact that did not exist before he acted on it.⁶ Competition is the

³Ludwig von Mises, *Economic Calculation in the Socialist Commonwealth*, trans. S. Adler (Auburn, AL: Ludwig von Mises Institute, 1990).

⁴Mises, *Human Action*, 758–759.

⁵Friedrich A. Hayek, "The Use of Knowledge in Society," *American Economic Review* 35, no. 4 (1945): 519–530.

⁶Israel M. Kirzner, *Competition and Entrepreneurship* (Chicago: University of Chicago Press, 1973); and Jesús Huerta de Soto, *Socialism, Economic Calculation and Entrepreneurship*, trans. Melinda A. Stroup (Cheltenham: Edward Elgar, 2010).

procedure that generates this knowledge, and prices are its running output. The central planner thus faces yet another impossible task, procuring information that does not yet exist, since the discovering is performed by the very rivalrous activity that central direction displaces.

In each of these arguments, the diagnosis of an intervention is performed by the market itself, in public. The classical critiques establish that intervention must fail, but they rely on the assumption that what converts that failure into correction is that the failure is public, priced, and attributable to its cause. And herein lies the problem: if this implicit assumption is violated, the intervention still fails, except there is no means by which policymakers can observe its failure.

What Programmable Money Changes

Rothbard provides a typology of state intervention, distinguishing three kinds: autistic, binary, and triangular, although all three share the crucial similarity of operating on actions, in that the object of each decree is a deed.⁷ The state forbids, compels, or taxes something a person does, and enforcement is therefore necessary only after the fact, since the act must first be detected, attributed, and then prosecuted. The gap between the decree and its enforcement is where the error signal lives. A prohibited exchange that happens anyway manifests as a black-market price, and a censored newspaper becomes a known and nameable act of suppression.

Programmable money interposes between state intervention and the otherwise ensuing signal. A central bank digital currency whose protocol declines a payment does not prohibit the exchange on paper and then, when the exchange occurs anyway, enforce its regulation; instead, it fundamentally renders the exchange inexpressible. Lessig's formula that "code is law" understates the difference, since law forbids and must then be enforced, whereas code circumscribes the space of possible action in advance.⁸ The result is not an expansion of the Rothbardian typology, because his types classify decrees by the nature of the state action. Programmable money is an intervention of an entirely foreign dimension: intervention in the *conditions* of action.

Its cost is counterfactual. What programmable money destroys is the entrepreneurial discoveries never made, because the actions that would have occasioned them were preempted at the level of the protocol. Since that knowledge originates in the very act of discovery,

⁷Murray N. Rothbard, *Man, Economy, and State with Power and Market*, Scholar's ed. (Auburn, AL: Ludwig von Mises Institute, 2009), 1057–1058.

⁸Lawrence Lessig, *Code: Version 2.0* (New York: Basic Books, 2006).

programmable money does not merely conceal information from the planner; it prevents the information from ever coming into existence. Bastiat's "unseen" was, in his account, a failing of the observer that the good economist trains himself to overcome;⁹ programmable money places the unseen beyond the reach of that training, since the foreclosed exchanges generate no data at all. A transaction blocked by a programmable wallet produces no black-market price, no queue, no photograph of empty shelves, only a private error message dispersed among individuals whom no market gathers up, while the exchanges deterred from ever being attempted produce nothing at all.

This is not immediately apparent. It might seem at first that digital infrastructure solves Hayek's epistemic problem. A system that underlies every exchange appears to hold more information than any price system could convey, so that comprehensive surveillance would seem to dissolve rather than deepen the knowledge problem. The thought mistakes the kind of knowledge at issue. Transaction logs record what people did inside an administered choice set; the knowledge that matters concerns what they would have valued and discovered under the alternatives the protocol forecloses, and one cannot datamine a counterfactual. A subtler version notes that programmable money logs its own refusals, so the planner sees frustrated demand directly. But refusal logs decay as the control binds: once a rule is learned, attempts cease, and the record is fullest where the intervention matters least. An attempt is in any case not a price: it registers that something was wanted, not how intensely or at what rate of exchange. And because discovery is sequential,¹⁰ the plans foreclosed by a single blocked first step never appear as attempts at all. Total surveillance of an administered system yields complete information about the administration and almost none about the economy it has displaced. The calculation problem is not solved by programmable money; it is enforced by it.

Detection and Restraint

Two consequences thus follow. First is *feedback substitution*. Without price signals, the planner must then govern via compliance metrics; but these metrics only measure enforcement, not coordination, and by Goodhart's law they fail as accurate proxies once they become the goal

⁹Frédéric Bastiat, *Ce qu'on voit et ce qu'on ne voit pas* [That Which Is Seen and That Which Is Not Seen] (Paris, 1850).

¹⁰Kirzner, *Competition and Entrepreneurship*.

of the planner's optimization.¹¹ The deeper problem is that near-total compliance looks identical whether the system is actually coordinating activity or discreetly undermining discovery. This is especially worrying since perfect compliance is most easily achieved precisely where the most unconventional and, therefore, most valuable entrepreneurial conduct has been preempted, because entrepreneurship is by its nature a departure from established patterns, and a protocol calibrated for the familiar will naturally be far more likely to flag or decline the unfamiliar.

But suppose the epistemic problem is somehow resolved and that the planner can still perceive failure. A second danger remains that is independent of the first: *the collapse of the marginal cost of intervening*. Under analog intervention, each round of detection and correction demands new legislation, inspectors, and prisons, an apparatus that is itself visible, costly, and contestable. Under programmable money, a change in the currency's configuration achieves the same ends at negligible cost and with greater reach. The European Central Bank has proposed individual holding limits for the digital euro of roughly three to four thousand euros¹² and has pledged that it would never be programmable money.¹³ Whatever the sincerity of that pledge, the holding limit is a few lines of mutable code, and a promise of non-programmability is a promise about a capability that is nonetheless retained. The difficulty is one of time inconsistency:¹⁴ a state that can re-parameterize its money at no cost cannot credibly bind itself not to, since every future crisis (capital flight, a pandemic, a fiscal emergency) supplies a fresh occasion to renege. Nor do legal safeguards close the gap, for judicial review is reactive: it needs a visible, attributable act to fasten on. Canada's freezes could be litigated because they were discrete, public, and traceable to a named decree; parameter changes that surface only as private payment failures present no such act, produce no plaintiff, and leave no record, and even a successful challenge arrives years too late. A rule forbidding the use of an existing capability is structurally weaker than the absence of the capability, since every emergency renegotiates the former, but none can conjure away the latter.

¹¹Charles A. E. Goodhart, "Problems of Monetary Management: The U.K. Experience," in *Papers in Monetary Economics*, vol. 1 (Sydney: Reserve Bank of Australia, 1975).

¹²Fabio Panetta, "The Digital Euro and the Evolution of the Financial System" (introductory statement to the Committee on Economic and Monetary Affairs of the European Parliament, June 15, 2022).

¹³European Commission, *Proposal for a Regulation on the Establishment of the Digital Euro*, COM(2023) 369 final (Brussels, 2023).

¹⁴Finn E. Kydland and Edward C. Prescott, "Rules Rather than Discretion: The Inconsistency of Optimal Plans," *Journal of Political Economy* 85, no. 3 (1977): 473–492.

The dynamic is clearest in a counterfactual. In February 2022, the Canadian government invoked the Emergencies Act to freeze roughly 280 accounts and other financial products tied to the convoy protests, without court orders.¹⁵ The episode produced exactly the error signals the classical account predicts: the banks had to be conscripted; the freezes drew international attention; a mandatory inquiry followed; and the Federal Court found the invocation unreasonable and the measures contrary to the Charter.¹⁶ The Federal Court of Appeal upheld that ruling in January 2026,¹⁷ and the federal government has since sought leave to appeal to the Supreme Court.¹⁸ The corrections arrived two and four years after the freezes and remain contested. Now suppose Canada were running a mature retail CBDC with identity-linked wallets. The same freezes would require no emergency statute, no bank conscription, and no public declaration; they would be routine flags in a compliance system, applied automatically, perhaps without the targets ever learning why their payments failed. One might object that people would notice and that journalists or advocacy groups would assemble the pattern even if no market did. Two features resist that assembly. A declined payment is individually deniable: it reaches the person affected as a glitch, a fraud flag, or an ordinary shortfall, so that even many collected complaints cannot show they share a single deliberate cause, not without the rule set itself. A freeze under the Emergencies Act, by contrast, was a single, public, nameable act, attributable to one decree at one moment, and that is what gathered scattered grievances into an inquiry and the rulings that followed. And aggregation can in any case only reach events that occurred, whereas the deepest losses here are the exchanges deterred before they are ever attempted, which leave no complainant to interview. The signal that generated two court rulings would never fire. Canada 2022 is best read not as a digital-age scandal but as the last generation of intervention old-fashioned enough to be caught.

What programmable money entails, then, is the Misesian cycle with its brakes removed. Distortions accumulate as foregone discoveries rather than visible crises; the planner reports success, and the political reckoning through which interventions have historically been rolled back never arrives.

¹⁵Public Order Emergency Commission, *Report of the Public Inquiry into the 2022 Public Order Emergency* (Ottawa, 2023).

¹⁶*Canadian Frontline Nurses v. Canada (Attorney General)*, 2024 FC 42.

¹⁷*Canada (Attorney General) v. Canadian Civil Liberties Association*, 2026 FCA 6.

¹⁸“Federal Government Appeals Emergencies Act Ruling to Supreme Court,” CBC News, 2026.

The eNaira and the Limits of Exit

The natural rejoinder is that this overstates the planner's reach, since people can always choose not to use digital currencies. Nigeria's eNaira, the first African central bank digital currency, is the standing example: a year after launch, 98.5 percent of wallets sat unused in any given week.¹⁹ This seems to immediately circumvent the empirical problem that programmable money poses. If programmable money's non-adoption is itself a visible signal, then its use seems far less likely to become prevalent. I do not deny this. The emptiness of the eNaira is a demonstrated preference against programmable money, expressed through the use of substitutes like cash, mobile money, and cryptocurrency. In short, this visibility does not hinder the Misesian cycle of progression.

However, some notable differences can be elucidated between the signal which non-adoption of programmable money produces and other typically produced signals. The first concerns resolution and attributability of the non-adoption signal. A black-market price is able to report how severely a control affects consumers, where it bites, whom exactly it affects, and how these effects change over time. An unused-wallet statistic reports only that people declined the system; it cannot say who the system foreclosed, or which transactions, and of what size, were foreclosed. What's more, the signal is also not attributable in the way a shortage is attributable to its price ceiling. The realm of possible interpretations of policy failure is, at least in the eyes of the planner, vast, and thus the planner very well might read non-adoption as a marketing or education failure, which prescribes extending the system rather than questioning it. In fact, this was what the IMF, having documented the non-adoption, recommended as a policy drive to mitigate the lackluster response in Nigeria.²⁰

When persuasion doesn't work, however, the planner may turn to more coercive means. In December 2022 the Central Bank of Nigeria engineered a nationwide cash shortage by demonetizing old notes and failing to supply adequate replacements, producing protests and riots. The CBN's governor explicitly named the destination as the fully cashless economy.²¹ When such a system fails, the signal is audible only so long as the substitutes remain alive, so

¹⁹J. Ree, *Nigeria's eNaira, One Year After*, IMF Working Paper No. 2023/104 (Washington, DC: International Monetary Fund, 2023).

²⁰Ree, *Nigeria's eNaira*.

²¹Human Rights Foundation, "CBDC Tracker: Nigeria," accessed June 2026, <https://cbdctracker.hrf.org>.

throttling them becomes the logical next step for a planner unknowingly perpetuating the Misesian cycle.

Visible Failure as a Public Good

One contention remains, since control of code is not always a monopoly of the state. Private platforms routinely employ algorithms to shape consumer behavior, payment processors deplatform when needed, and app stores gate entry. All of this is done through code, whose control is easily centralized and leveraged on a daily basis. So it seems that programmable money would merely be a banal addition to societies already inundated with digital systems that resemble its capabilities. In other words, if code were what suppressed error signals, the argument would prove too much and indict the market's own, already ubiquitous infrastructure rather than government intervention specifically.

In fact the argument does not fail in this manner, because code *qua* code is not the issue; rather, it is the context in which it is employed which engenders programmable money's unique outcomes. Private code is embedded in the competitive profit-and-loss system. A platform whose code destroys value for its users cedes engagement, advertisers, and capital to its rivals, and then must answer for said losses in front of fickle investors.²² The state, however, faces no such test. It is financed by taxation rather than investment, provides public goods which are collectively paid for through binary interventions, and may lawfully suppress its competitors.²³ And the empirical relation between the private sector and the state can often be an autistic one, imposed by the latter upon the former: in Canada, the state did not construct the apparatus that allowed it to freeze the workers' funds; it commandeered the banks'.

Ultimately, if the visible failure of intervention is what has historically allowed for the cessation of further government intervention, then visible failure is itself an economic good. And the instruments that produce it, like cash, permissionless payment rails, and monies the state does not control, are, as a result, not relics from a pre-digital economy. The value of such instruments is not in the privacy or convenience they afford, real as those benefits are; on an institutional level, it is that they provide the ability to facilitate discovery of mismanagement in an economy.

²²Mises, *Human Action*.

²³Hans-Hermann Hoppe, *A Theory of Socialism and Capitalism* (Auburn, AL: Ludwig von Mises Institute, 2010), 183–186; and Rothbard, *Man, Economy, and State*, 948.

Competition among monetary infrastructures is itself a discovery procedure in Hayek's sense,²⁴ one that programmable money precludes.

Programmable money cannot save the planner, for it supplies none of the prices he lacks; worse, it deprives everyone else of the failures that price information would have revealed. A society that surrenders visible failure does not thereby simply abolish the failures of its monetary policy. It abolishes the chance for the failures to be seen.

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